

# GTSC2143 Machine Learning for Business

## Tutorial 7

Please write down your answers in this document and submit it at iSpace by the end of this tutorial.

### Activity 1. Data Loading and Preprocessing

#### 1. Load the Dataset

```
import pandas as pd

# Load the customer churn dataset

data = pd.read_csv("https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv", encoding='iso8859-1')
```

#### 2. Explore the Dataset:

- Display the data shape
- Show the column names
- Show first 5 rows
- Check for missing values and handle any missing values appropriately

#### 3. Analyze Class Distribution

- Examine the distribution of the target variable (Churn) and calculate the class counts and percentages.
- Create a bar chart to visualize the class distribution.

### Activity 2. Data Processing

#### 1. Feature Preprocessing

- Encode categorical variables using appropriate techniques
- Scale numerical features using `StandardScaler`
- Split the data into training (80%) and testing (20%) sets using `random_state=42`

#### 2. Baseline Model Performance

- Train a logistic regression model on the original (imbalanced) training data
- Evaluate the model on the test set using confusion matrix
- Analysis: Which class is being predicted poorly? Why might accuracy be misleading for this dataset?

### Activity 3. SMOTE for Class Balance

#### 1. Apply SMOTE

- Import SMOTE from `imblearn` and apply it to the training data only
- Compare the class distribution before and after SMOTE
- Explain why SMOTE should only be applied to training data

#### 2. Model Comparison

- Train a logistic regression model on the SMOTE-balanced data
- Evaluate and compare the performance with the baseline model
- Discuss the trade-offs between precision and recall. Which approach is better for customer churn prediction?

#### **Activity 4. Linear vs Non-Linear Classifiers**

##### **1. Linear Classifiers**

Train and evaluate the following linear classifiers on SMOTE-balanced data:

- a) Linear SVM (SVC(kernel='linear'))
- b) Gaussian Naive Bayes

##### **2. Non-linear Classifiers**

Train and evaluate the following non-linear classifiers on SMOTE-balanced data:

- a) Random Forest (n\_estimators=100)
- b) SVM with RBF kernel (SVC(kernel='rbf'))
- c) Decision Tree (max\_depth=10)

##### **3. Model Comparison**

- a) Create a comparison table showing accuracy, precision, recall, and F1-score for all classifiers.
- b) Generate ROC curves for all classifiers on the same plot.

##### **4. Feature Importance Analysis**

- a) Extract and display the top 10 most important features from the Random Forest model
- b) Analyze the coefficients from the Logistic Regression model
- c) Based on performance metrics, select the best model.

- End of Tutorial 7 -